A MAJOR PROJECT REPORT

ON

STRATEGIC APPLICATION OF ARTIFICIAL INTELLIGENCE IN CAD

Automated design, verification and drafting of standard machine joints from a plain language request

Submitted in partial fulfilment of the requirements
for the award of the degree of

Bachelor of Engineering
in
Mechanical Engineering

Submitted by

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Under the guidance of

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Session 20__ - 20__

CERTIFICATE

This is to certify that the major project report entitled **"Strategic Application of Artificial Intelligence in CAD"** is a bona fide record of the work carried out by the students named below, of the final year Bachelor of Engineering course in Mechanical Engineering, in partial fulfilment of the requirements for the award of the degree during the session 20__ - 20__.

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The work presented in this report has been carried out under my supervision and guidance. To the best of my knowledge, the content of this report has not been submitted elsewhere for the award of any other degree or diploma.

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DECLARATION

We hereby declare that the major project report entitled **"Strategic Application of Artificial Intelligence in CAD"**, submitted to the Department of Mechanical Engineering, Shri Govindram Seksaria Institute of Technology and Science, Indore, Madhya Pradesh, is an authentic record of our own work carried out during the session 20__ - 20__.

The design formulas and proportions used in this work are taken from published design literature, and every such source is cited in the body of the report and listed in the references. Where the software developed in this project departs from a cited source, the departure and its reason are stated explicitly in Chapter 5 rather than left implicit.

We further declare that the content of this report has not been submitted, in whole or in part, to any other institute or university for the award of any degree or diploma.

Signatures of the candidates

Date and Place

ACKNOWLEDGEMENT

We express our sincere gratitude to _____, _____, Department of Mechanical Engineering, for the guidance and encouragement given throughout this work. The direction to keep every computed dimension traceable to a published formula, rather than to a convenient result, shaped the project more than any other single input.

We thank **Dr. B. R. Rawal**, Head of the Department of Mechanical Engineering, for providing the facilities and the academic environment in which this work was carried out, and the faculty of the department for their suggestions during the review presentations.

We acknowledge the National Programme on Technology Enhanced Learning and the Indian Institute of Technology Kharagpur, whose openly published course material on the design of machine elements formed the primary reference for this work, and the Internet Archive, through which the out of copyright texts by Unwin and Leutwiler were obtained.

Finally, we thank our families and classmates for their support during the course of this project.

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ABSTRACT

Designing a standard machine joint is a repetitive calculation. For a knuckle joint the designer computes the rod diameter from tension, sizes the pin separately for shear and for bending, takes the larger of the two, derives the remaining proportions from the rod diameter, and then verifies nine distinct failure modes against allowable stresses. Changing the load requires the entire sequence to be repeated. The drafting that follows is mechanical but unavoidable.

This project presents a software system that performs that sequence automatically from a request written in ordinary English. The user types a sentence such as *knuckle joint for 50 kN*. A language model extracts the parameters from the sentence. All subsequent work is deterministic: the joint is sized from standard design formulas, every failure mode is checked against the allowable stress, any dimension that fails is increased until it passes, each part is generated as an exact solid model, and a first angle A3 drawing sheet is produced with dimensions and a title block.

Six joints are implemented: the knuckle joint, the socket and spigot cotter joint, the sleeve and cotter joint, the gib and cotter joint, the hex bolted joint, and the protected type rigid flange coupling. Together they apply 44 distinct failure checks. Every proportion and every check has been compared against published design literature, and the comparison identified three errors in the sources themselves, including a worked example whose own answer fails its own check by a margin of 23 percent. The system is covered by 290 automated tests and is deployed as a public web application.

The central design decision is that the language model reads the request only. It selects no dimension and performs no check. Interpreting a phrase is a language problem; selecting a pin diameter is an engineering problem with a correct answer, and that remains with deterministic code whose every step can be traced to a printed page.

Keywords: computer aided design, machine design, knuckle joint, cotter joint, flange coupling, design automation, parametric modelling, natural language interface, finite element analysis, first angle projection.

TABLE OF CONTENTS

Certificate ..... **i**

Declaration ..... **ii**

Acknowledgement ..... **iii**

Abstract ..... **iv**

List of Figures ..... **vi**

List of Tables ..... **vii**

1. Introduction **1**

1.1 Background 1

1.2 Statement of the problem 1

1.3 Objectives 2

1.4 Scope and limitations 2

1.5 Organisation of the report 2

2. Literature Review **3**

2.1 Design of cotter and knuckle joints 3

2.2 Design of shaft couplings 3

2.3 Existing approaches to design automation 4

2.4 Gap identified 4

3. Methodology **5**

3.1 Overall approach 5

3.2 Role of the language model 5

3.3 Sizing procedure 6

3.4 Verification of failure modes 6

3.5 Geometry and drawing generation 7

3.6 Finite element cross check 7

3.7 Software tools used 8

4. Design and Implementation **9**

4.1 System architecture 9

4.2 Joints implemented 9

4.3 Worked example: knuckle joint for 50 kN 10

4.4 Drawing sheet generation 12

4.5 User interface and deployment 12

5. Results and Discussion **13**

5.1 Design results 13

5.2 Verification against published literature 13

5.3 Discrepancies found in the sources 14

5.4 Finite element comparison 15

5.5 Software verification 15

6. Conclusion and Future Scope **16**

References **17**

Appendix A: Screenshots of the working system **18**

LIST OF FIGURES

Fig. 1: First angle A3 drawing sheet generated for the knuckle joint of Section 4.3

Fig. 2: Assembled model of the knuckle joint, rotatable in the browser, also available exploded and coloured by utilisation

Fig. 3: Working printed for a single run, showing the computed values before rounding, the rule and source of each dimension, and every chec

Fig. 4: The request screen. A requirement is typed in ordinary English, or one of the example requests is selected

Fig. 5: Parameters extracted from the request, presented for confirmation or correction before the design proceeds

Fig. 6: Every computed dimension with the rule that produced it and its source; a plain language explanation is available for each

Fig. 7: Every failure check with its substituted arithmetic, its allowable stress and its utilisation

Fig. 8: Components coloured by utilisation, so that the member nearest its limit is apparent without reading the table

Fig. 9: Files produced by a single run: a STEP file for each component, the assembly, the drawing sheet, the DXF, the FreeCAD file and a wri

LIST OF TABLES

Table 1: Software tools used in the implementation

Table 2: The six joints the software designs, with the duty each was sized for

Table 3: Governing calculations before rounding to a preferred size

Table 4: Dimensions computed for a knuckle joint carrying 50 kN, with the rule behind each

Table 5: Failure checks for the same joint, computed at the actual load of 50 kN

Table 6: Verification of each joint against published design literature

CHAPTER 1 INTRODUCTION

1.1 Background

A machine joint transmits load between two members. The standard joints taught in machine design, the knuckle joint, the cotter joint and the shaft coupling among them, are not designed by judgement. Each is sized by a fixed sequence of formulas and then verified against a fixed list of failure modes. The sequence is well established and appears in essentially the same form in every standard text on the subject.

Because the sequence is fixed, it is repetitive. A change in the applied load invalidates every dimension and requires the whole calculation again. In academic practice this repetition consumes the majority of the time spent on a design problem, while the engineering judgement involved, choosing allowable stresses and interpreting the result, occupies very little of it. The drafting that follows the calculation is similarly mechanical.

Computer aided design software removes the labour of drawing but not the labour of calculation. A modeller will faithfully produce whatever dimensions it is given and will not object if a pin is too slender to carry the bending moment placed on it. The calculation remains the designer's responsibility and remains manual.

1.2 Statement of the problem

The problem addressed by this project is the gap between stating a requirement and obtaining a verified design. At present a designer must translate a requirement such as *a knuckle joint to carry 50 kN* into a sequence of formulas by hand, evaluate them, check the result against several failure modes, iterate where a check fails, and only then begin modelling and drafting.

A system that accepts the requirement as written and returns a verified design, complete with solid models and a drawing sheet, would remove the repetition without removing the engineering. The difficulty is that the interpretation of a written requirement is not a calculation, and the calculation must not be delegated to a system that cannot be checked.

1.3 Objectives

1. To develop a system that accepts a design requirement stated in ordinary English and extracts the joint type, the applied load and the allowable stresses from it.
2. To size the joint from established design formulas and to verify every applicable failure mode against the allowable stress, iterating where a check is not satisfied.
3. To generate exact solid models of every component and an assembled model in a neutral exchange format that opens in commercial CAD software.
4. To generate a first angle projection drawing sheet with dimensions and a title block conforming to Indian drawing practice.
5. To verify every proportion and every failure check used by the system against published design literature, and to record any disagreement with a source explicitly.
6. To make the system usable through a web browser without any installation.

1.4 Scope and limitations

The scope of this work is limited to six standard joints under static loading. Fatigue loading, stress concentration factors, tolerances and fits, and manufacturing process selection are outside the scope. The finite element facility included in the system performs a single linear static analysis on one component and is intended as an independent cross check on the formula, not as a substitute for a complete analysis.

1.5 Organisation of the report

Chapter 2 reviews the design literature consulted and the existing approaches to design automation. Chapter 3 describes the methodology, including the division of responsibility between the language model and the deterministic calculation. Chapter 4 describes the implementation and presents a complete worked example. Chapter 5 presents the results, the verification against published sources, and the discrepancies found in those sources. Chapter 6 concludes and identifies further work.

CHAPTER 2 LITERATURE REVIEW

2.1 Design of cotter and knuckle joints

The primary reference used in this work is the course material on the Design of Machine Elements published by the Indian Institute of Technology Kharagpur through the National Programme on Technology Enhanced Learning. Module 4 of that course treats pin joints, keys and threaded fasteners. Lesson 2 gives the complete design procedure for the socket and spigot cotter joint and for the knuckle joint, including the empirical proportions expressed as multiples of the rod diameter and the full list of failure modes to be verified.

Two further texts were used, both out of copyright and obtained from the Internet Archive. Unwin, *The Elements of Machine Design* (1899), treats the gib and cotter joint on page 179 and gives the division of the slot breadth between cotter and gib, the thickness as a fraction of that breadth, and the permissible range of taper. Leutwiler, *Elements of Machine Design* (1917), gives in Article 98 a list of ten distinct ways in which a cotter joint may fail, together with the expression governing each.

The availability of two independent treatments of the cotter joint, published eighteen years apart, proved significant. Where the two sources agree, the expression used in this work can be regarded as settled. Where they disagree, the disagreement itself is a finding, and is reported in Section 5.3.

2.2 Design of shaft couplings

Leutwiler, Article 275, gives the analysis of the flange coupling as three groups of failure: failure of the key by shear or by crushing, failure of the coupling bolts by shear or by crushing, and shearing of the flange at the hub. The expressions given there for the bolt diameter are used in this work. The same article records that no uniform set of proportions for flange couplings had been agreed as a standard, and that the published proportion tables represent the practice of individual manufacturers. This observation directly limits what can be claimed about the proportions used in the present work and is reflected in Section 5.2.

The two key equations, shear across the whole key and bearing across half the key thickness, are taken from NPTEL Module 4 Lesson 1.

2.3 Existing approaches to design automation

Parametric modelling is standard in commercial CAD software. A model driven by a table of parameters will regenerate when those parameters change, and this removes the drafting labour associated with a change in size. It does not, however, determine the parameters. The designer must still compute them, and the software will regenerate an unsafe model as readily as a safe one.

Recent work applying large language models to CAD has largely taken the form of generating modelling code directly from a description. Such systems produce geometry quickly but the dimensions they produce cannot be traced to a design calculation, and there is no mechanism by which a failure mode is verified. For an academic or professional design task, geometry that cannot be justified is of limited value.

2.4 Gap identified

The gap is therefore not in modelling and not in language interpretation, both of which are mature, but in the connection between them. What is missing is a system in which a written requirement is interpreted by a language model, the resulting parameters drive a deterministic design calculation taken from published literature, and the calculation itself remains open to inspection. The present work addresses that gap.

CHAPTER 3 METHODOLOGY

3.1 Overall approach

The system is organised as five stages executed in sequence. The written request is interpreted into parameters; the joint is sized; every failure mode is verified; the geometry is generated; and the drawing sheet is produced. Each stage passes a definite result to the next, and the intermediate results of every stage are reported to the user.

3.2 Role of the language model

The language model is used at the first stage only. It receives the written request and returns the joint type, the applied load in newtons, and any allowable stresses stated. It selects no dimension, applies no proportion and performs no check.

This division is deliberate and is the principal design decision of the project. Determining that the phrase *50 kN* denotes a load of 50,000 N, and that *socket and spigot* names the cotter joint, is a problem of language interpretation, at which such a model is effective. Determining the diameter of the pin is a problem of engineering with a single correct answer obtainable from a formula, and it is therefore performed by deterministic code whose every step is traceable to a published page.

A safeguard is applied to the model output. Any numerical value returned by the model that does not appear in the original request is discarded, and the user is informed. This prevents a plausible but invented figure from entering the calculation. Where no language model is available, the system falls back to pattern matching rules and states which interpreter was used.

3.3 Sizing procedure

The governing dimension is computed first from the primary failure mode. For a knuckle joint this is the rod diameter obtained from direct tension. The computed value is then rounded up to the next standard preferred size rather than used as computed, in accordance with the practice given in the source material.

Remaining dimensions follow from the empirical proportions given in the source, expressed as multiples of the governing dimension. Where a component has an independent failure mode, as the pin of a knuckle joint has in both shear and bending, that component is sized separately from each applicable expression and the larger requirement is adopted.

A design factor of 1.25 is applied. The joint is sized for 1.25 times the stated load and the resulting stresses are then reported at the actual load. The consequence is that the governing check reports approximately 75 to 80 percent of the allowable stress rather than 100 percent, leaving margin in hand. Sizing exactly to the allowable produces a design that satisfies every check on paper while retaining nothing for material variation or overload.

3.4 Verification of failure modes

Every applicable failure mode is evaluated and compared with the corresponding allowable stress. Where a check is not satisfied, the dimension responsible is increased by one millimetre and the entire check set is re-evaluated. The process repeats until all checks are satisfied. The dimension that was increased, and the check that caused it, are both reported.

3.5 Geometry and drawing generation

Each component is generated as an exact solid using a parametric modelling kernel, and is written in the STEP neutral exchange format, which opens in Siemens NX, SolidWorks, FreeCAD, Onshape and Fusion. The assembly is written as a further STEP file with each component in its assembled position.

The drawing sheet is produced by driving FreeCAD without its graphical interface. The assembled geometry is projected to obtain the front view, the side view and the plan, arranged in first angle projection in accordance with Indian practice, with an isometric view, dimension lines and a title block. The same views are also written in DXF format.

3.6 Finite element cross check

An optional finite element analysis is provided as an independent check on the calculated stress. One component is meshed and solved as a linear static problem using the CalculiX solver, and the resulting peak stress is compared with the value obtained from the formula.

Where the computed peak exceeds the calculated value by more than half, the analysis is repeated on a mesh of half the element size. If the peak then moves by more than five percent the result is reported as not converged, since a peak that continues to change with mesh refinement indicates a stress concentration at a sharp corner or at a constrained face rather than a stress that may be designed to.

3.7 Software tools used

Table 1 lists the software used to implement the system. All of it other than the language model is open source.

Table 1: Software tools used in the implementation

Function	Tool	Remark
Solid modelling	CadQuery 2.8 on the OCCT kernel	Open source, scripted parametric modelling
Drawing and DXF	FreeCAD, headless	Projection and title block
Finite element analysis	CalculiX with gmsh	Linear static
Language interpretation	Claude language model	Parameter extraction only
Application	Python standard library	No external framework
Deployment	Container hosted on Vercel	Public, no installation

CHAPTER 4 DESIGN AND IMPLEMENTATION

4.1 System architecture

The design rules for each joint are held as data rather than as code. Each joint is described by a file recording its empirical proportions, the source of each proportion, the default allowable stresses and the preferred size series. The calculation code reads these files. A consequence of this arrangement is that the source of any dimension can be reported to the user at the time it is computed, and a correction to a proportion is a change to data rather than to code.

4.2 Joints implemented

Six joints are implemented. Table 2 records the number of components, the number of dimensions computed and the number of failure checks applied to each, together with the utilisation of the busiest check at the duty shown.

Table 2: The six joints the software designs, with the duty each was sized for

Joint	Parts	Dimensions	Checks	Duty used for this table	Busiest check
Knuckle joint	5	12	9	Load P = 50,000 N	76 %
Socket and spigot cotter joint	3	14	10	Load P = 35,000 N	67 %
Sleeve and cotter joint	5	12	8	Load P = 50,000 N	77 %
Gib and cotter joint	4	20	8	Load P = 50,000 N	79 %
Hex bolt, nut and washer	3	11	2	Load P = 20,000 N	71 %
Protected type rigid flange coupling	12	15	7	Torque T = 500,000 N mm (500 N m)	79 %

The gib and cotter joint requires comment. Unwin gives the division of the breadth and the thickness but gives no proportions for the strap. Rather than adopt a ratio from another source or invent one, every dimension not given by Unwin is obtained by sizing it against a failure check until that check is satisfied. The result is that no dimension of that joint is unsupported: each is either taken from the source or derived from a stated failure mode.

4.3 Worked example: knuckle joint for 50 kN

The request *knuckle joint for 50 kN* is interpreted as the knuckle joint carrying an axial load of 50,000 N, with the default allowable stresses of 75 MPa in tension, 60 MPa in shear and 150 MPa in crushing. Applying the design factor, the joint is sized for 62,500 N.

The rod diameter follows from direct tension. The pin is then sized twice, once from double shear and once from bending, and the larger requirement governs. The three computed values before rounding are given in Table 3.

Table 3: Governing calculations before rounding to a preferred size

Expression	Computed (mm)	Effect
Rod in direct tension	32.57	Rod diameter adopted as 35 mm
Pin in double shear	25.75	Not governing, smaller than the rod
Pin in bending	43.72	Governs; pin diameter adopted as 45 mm

It is worth noting that the pin requirement from bending is nearly twice that from shear. A pin sized from shear alone would satisfy that check and bend in service. This is the reason the larger of the two requirements is adopted, and the source material takes the same step.

The remaining dimensions follow from the empirical proportions, each expressed as a multiple of the rod diameter. The complete set, with the rule and the source behind every value, is given in Table 4.

Table 4: Dimensions computed for a knuckle joint carrying 50 kN, with the rule behind each

Symbol	Meaning	Value (mm)	Rule and source
D	rod diameter	35	rod in tension: D = sqrt(4P / (pi sigma_t)), rounded up to a preferred size
d	pin diameter	45	governed by pin bending
d2	eye outer diameter	70	d2 = 2.0 x D, the rod diameter (outer diameter of eye) [NPTEL IIT KGP M4L2 fig 4.2.3.1]
d3	pin head and collar diameter	52.5	d3 = 1.5 x D, the rod diameter (pin head and collar diameter) [NPTEL IIT KGP M4L2 fig 4.2.3.1]
t	eye thickness	43.75	t = 1.25 x D, the rod diameter (thickness of single eye) [NPTEL IIT KGP M4L2 fig 4.2.3.1]
t1	fork plate thickness	26.25	t1 = 0.75 x D, the rod diameter (thickness of each fork plate) [NPTEL IIT KGP M4L2 fig 4.2.3.1]
t2	pin head thickness	17.5	t2 = 0.5 x D, the rod diameter (thickness of pin head) [NPTEL IIT KGP M4L2 fig 4.2.3.1]
collar_t	collar thickness	22.5	-
taper_d	taper pin diameter (big end)	8.75	-
clearance	gap between eye and fork base	4.5	-

With all dimensions determined, the nine failure modes are evaluated at the actual load of 50,000 N. The results are given in Table 5. Every check is satisfied, and the busiest, shear of the eye, reaches 76 percent of its allowable stress.

Table 5: Failure checks for the same joint, computed at the actual load of 50 kN

Failure mode	Expression used	Stress (MPa)	Allowable (MPa)	Utilisation	Result
rod tension	4P / (pi D^2)	52	75	69 %	Safe
pin double shear	P / (2 pi/4 d^2)	15.7	60	26 %	Safe
pin bending	32M / (pi d^3), M = P/2 (t1/3 + t/4)	55	75	73 %	Safe
eye tension	P / ((d2 - d) t)	45.7	75	61 %	Safe
eye shear	P / ((d2 - d) t)	45.7	60	76 %	Safe
eye crushing	P / (d t)	25.4	150	17 %	Safe
fork tension	P / ((d2 - d) 2 t1)	38.1	75	51 %	Safe
fork shear	P / ((d2 - d) 2 t1)	38.1	60	64 %	Safe
fork crushing	P / (d 2 t1)	21.2	150	14 %	Safe

4.4 Drawing sheet generation

The assembled geometry is projected to produce a first angle A3 sheet. Fig. 1 shows the sheet produced for the example above, at a scale of 1:2.5. The sheet carries the front view, the left side view, the plan and an isometric view, with dimension lines and a title block giving the load, the institution and the request from which the design was produced.

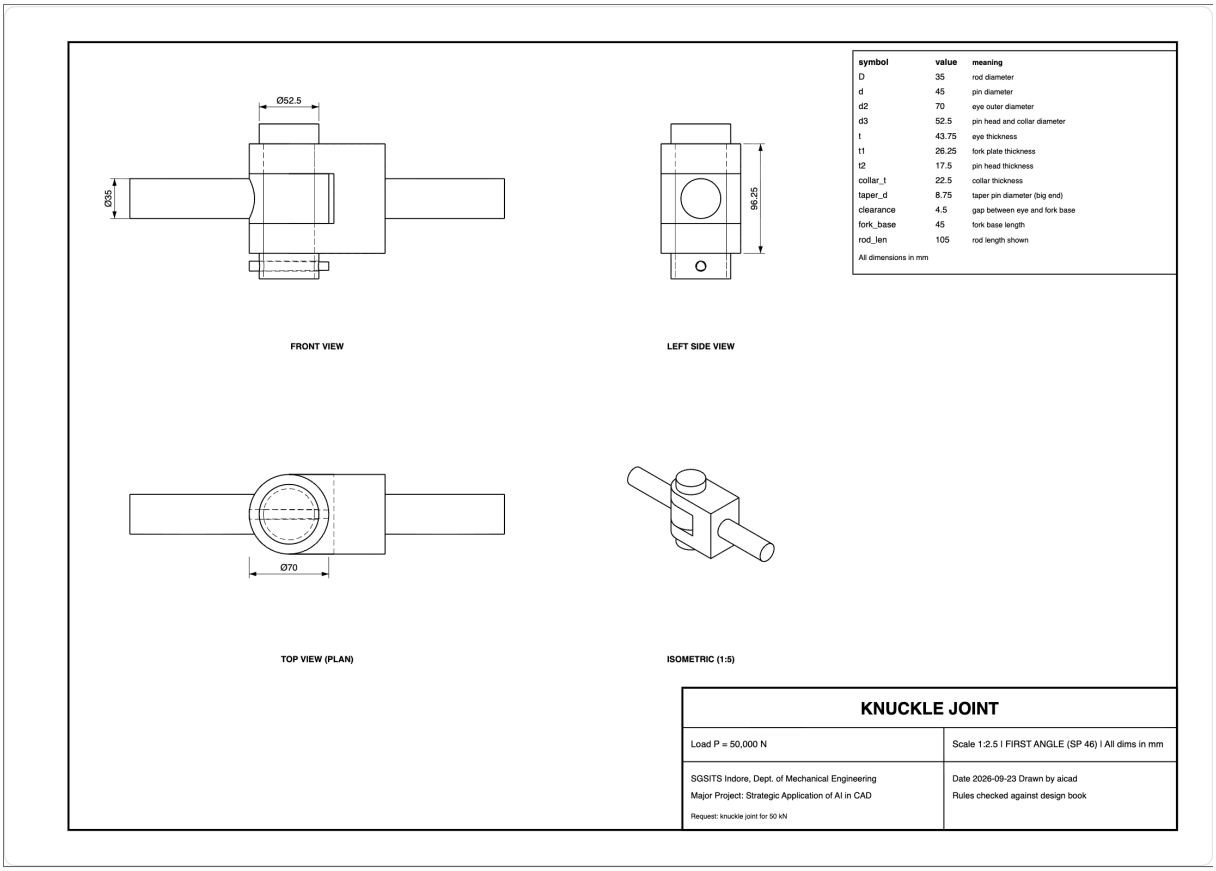


Fig. 1: First angle A3 drawing sheet generated for the knuckle joint of Section 4.3

4.5 User interface and deployment

The system is used through a web browser. The user enters the requirement, confirms or corrects the parameters extracted from it, and requests the build. The full working of the calculation is printed as it proceeds, and the results are presented as the 3D model, the drawing, the dimension list, the check list, a view colouring components by utilisation, and the generated files.

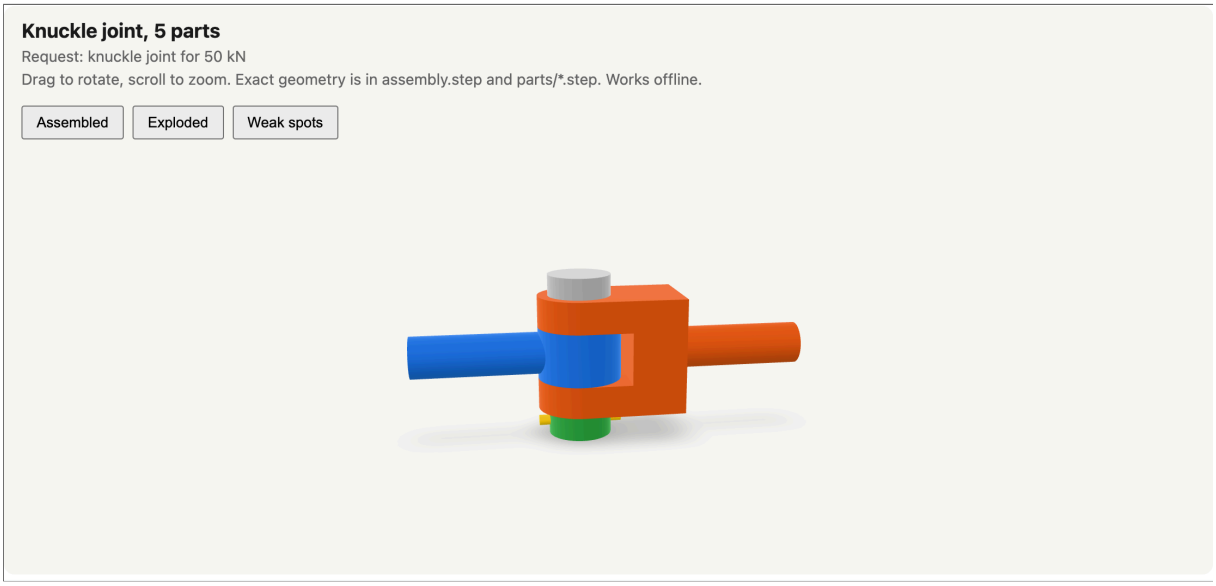


Fig. 2: Assembled model of the knuckle joint, rotatable in the browser, also available exploded and coloured by utilisation

CHAPTER 5 RESULTS AND DISCUSSION

5.1 Design results

All six joints size correctly, satisfy every applicable failure check, generate every component as a solid model and produce a drawing sheet. Each design completes in approximately ten seconds. The busiest check across the six joints lies between 67 and 79 percent of the allowable stress, which is the intended consequence of the design factor described in Section 3.3 and confirms that margin is retained in every case.

5.2 Verification against published literature

Every proportion and every failure check was compared against the sources reviewed in Chapter 2. The outcome is summarised in Table 6.

Table 6: Verification of each joint against published design literature

Joint	Source consulted	Outcome
Knuckle joint	NPTEL, IIT Kharagpur, Module 4 Lesson 2	Proportions and all nine checks verified
Socket and spigot cotter joint	NPTEL Module 4 Lesson 2; Leutwiler (1917), Art. 98	All ten checks verified against two independent sources
Gib and cotter joint	Unwin (1899), p. 179, Figs. 130 and 131	Breadth split, thickness and taper verified
Protected type flange coupling	NPTEL Module 4 Lesson 1; Leutwiler Art. 275	All seven checks verified twice; proportions not verifiable
Sleeve and cotter joint	No source found in four books searched	Checks verified by analogy; proportions unverified
Hex bolt and nut	NPTEL Module 4 Lesson 4	Tightening relation verified; size table unverified

Two entries require explanation. For the flange coupling, all seven failure checks are verified against two independent sources, but the empirical proportions are marked unverified. This is not an omission. As recorded in Section 2.2, Leutwiler states that no standard set of proportions was ever agreed, and that published tables reflect individual manufacturers' practice. The proportions used are therefore a convention and the system reports them as such. For the sleeve and cotter joint, four sources were searched and none treats that joint by name; its checks are verified by direct correspondence with the socket and spigot cotter joint, but its proportions remain unsupported and are reported as such on every run.

5.3 Discrepancies found in the sources

The verification exercise identified three errors in the published sources. They are recorded here because the system's behaviour deliberately differs from the printed material as a result.

1. A worked example that fails its own check.

The knuckle joint worked problem raises the pin diameter from 40 mm to 55 mm to satisfy bending, but subsequently evaluates the eye using the original 40 mm pin. Evaluated with the pin actually adopted, its own expression for shear of the eye gives 80 MPa against an allowable of 65 MPa, an exceedance of 23 percent. The present system instead enlarges the outer diameter of the eye and reports that it has done so.

2. An omitted failure mode.

The cotter joint procedure in the same source verifies crushing of the socket but does not verify crushing between the cotter and the spigot. Evaluated at the final answer given in that source, the omitted check gives 167 MPa against an allowable of 110 MPa. Leutwiler, Article 98, lists this mode as one of ten, and its expression is identical in form. The present system retains the check and sizes the cotter thicker in consequence. That two independent texts, published eighteen years apart, include a check which the lesson omits, is what establishes this as an omission rather than a matter of preference.

3. A constant printed incorrectly.

The initial tightening load for a bolted joint is printed as $284d$ kN. The worked example on the same page obtains 56.8 kN for an M20 bolt, which is consistent only with a constant of $2840d$ newtons. The printed relation is therefore in error by a factor of ten, and the source refutes itself. The system uses $2840d$ N and reports the tightening load on every bolted design, since at M20 it amounts to 56,800 N before any service load is applied.

5.4 Finite element comparison

The finite element facility was first validated against a case with a known answer. A bar of 20 mm diameter loaded in tension by 10 kN gives a computed peak of 32.22 MPa against the calculated 31.83 MPa, a difference of 1.2 percent, attributable to the constrained face.

Applied to the eye rod of the knuckle joint of Section 4.3, the computed peak is 57.9 MPa against the calculated 52.0 MPa, a difference of 11.5 percent. The excess is expected: the calculation treats the rod as a plain bar in tension and takes no account of the stress raised at the pin bore, whereas the finite element model does. Both values remain below the allowable stress of 75 MPa.

The convergence rule described in Section 3.6 proved necessary in practice. For one component the computed peak varied between 157 and 191 MPa across a sixteen fold increase in the number of nodes without settling, indicating a singularity at the constrained face rather than a physical stress. The system reports such a result as not converged and directs the user to the calculated value.

5.5 Software verification

The implementation is covered by 290 automated tests, executed on every change. These cover the sizing arithmetic, the generated geometry, the interpretation of requests, the drawing sheet and the finite element facility. Among them are tests which assert that the discrepancies recorded in Section 5.3 are genuine, so that the system cannot later be silently altered to reproduce the printed answers.

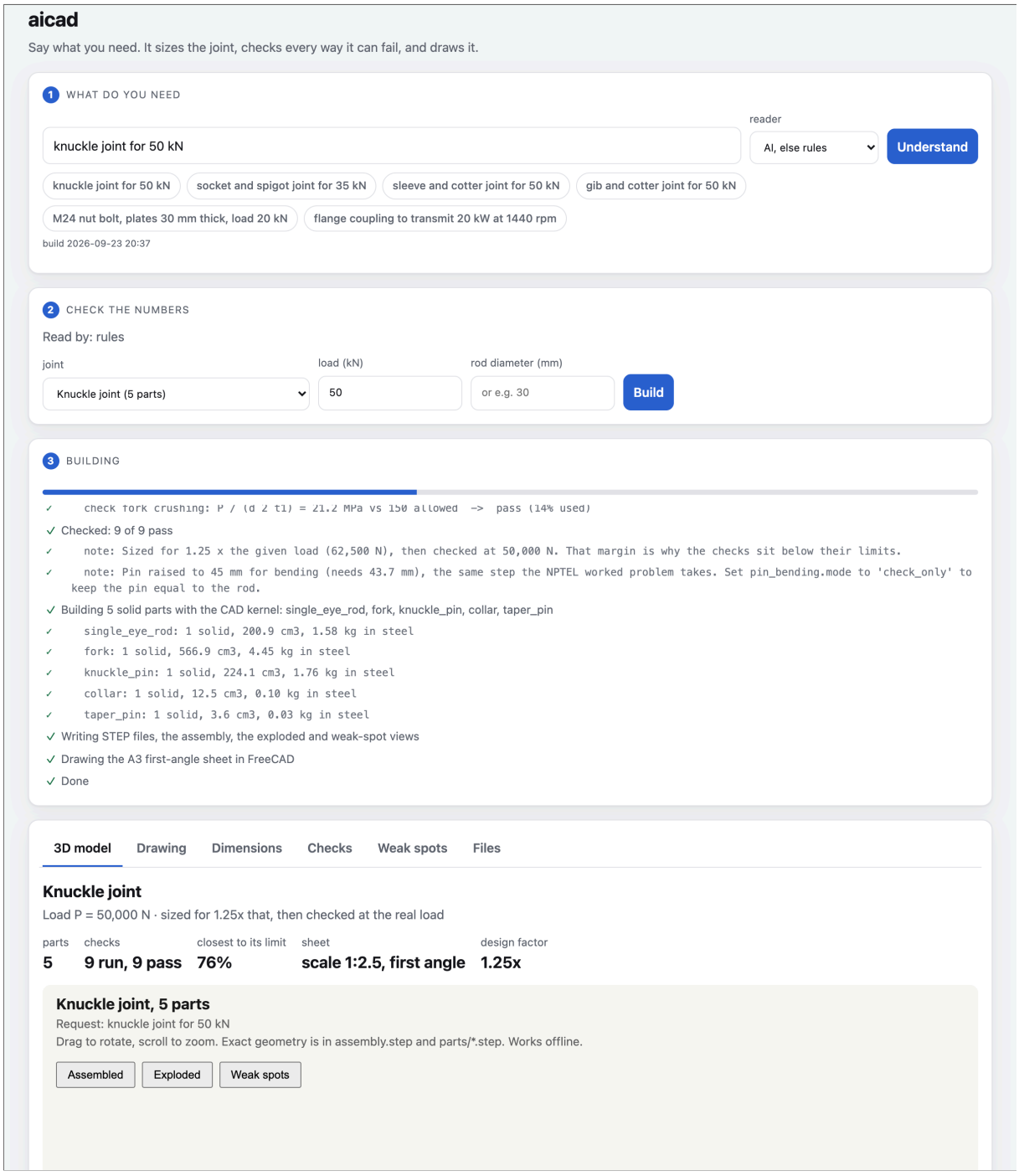


Fig. 3: Working printed for a single run, showing the computed values before rounding, the rule and source of each dimension, and every check with its substituted arithmetic

CHAPTER 6 CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

A system has been developed which accepts a machine joint requirement written in ordinary English and returns a complete design: every dimension computed from published design formulas, every applicable failure mode verified against the allowable stress, every component generated as an exact solid model, and a first angle drawing sheet produced with dimensions and a title block. Six joints are implemented, applying 44 distinct failure checks in total, and the system is covered by 290 automated tests.

The objective of keeping the design traceable has been met. Every dimension the system produces can be attributed either to a formula with a cited source or to an explicitly named failure check. The working of each run is printed in full, which allows the result to be verified rather than accepted.

The verification exercise against published literature produced an unanticipated result. Three errors were identified in the sources themselves, one of which is a worked example whose own answer exceeds its own allowable stress by 23 percent. This was found not by inspection but because the software evaluates every check on every design and reports the arithmetic, which makes an omitted or inconsistent check visible immediately.

The division of responsibility adopted here, in which the language model interprets the request and deterministic code performs the engineering, is the element of the work that is most generally applicable. It allows a natural language interface to be provided without any part of the design becoming unverifiable.

6.2 Future scope

1. **Additional joints.** Riveted and welded joints, treated in Module 10 of the source material, follow the same pattern and could be added without change to the architecture.
2. **Fatigue loading.** All sizing at present is static. Introducing an endurance limit and a stress concentration factor would extend the system to cyclically loaded joints.
3. **Tolerances and fits.** The drawing sheet carries nominal dimensions only. Adding fits for the pin and its bore would make the output directly usable for manufacture.
4. **Verification of the remaining proportions.** Two of the six joints have proportions without a supporting source. A design data handbook would settle both.
5. **Direct integration with commercial CAD.** Output is presently exchanged as STEP. A journal script for Siemens NX would allow the assembly to be constructed within the software rather than imported into it.
6. **Manufacturing outputs.** Material selection and cost estimation are partially implemented and could be extended to a bill of materials.

REFERENCES

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3. Leutwiler, O. A., *Elements of Machine Design*, McGraw Hill Book Company, New York, 1917. Article 98: analysis of a cotter joint. Article 275: analysis of a flange coupling.
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5. Bureau of Indian Standards, IS 2048: *Parallel Keys and Keyways*; IS 1363 and IS 1364: *Hexagon Head Bolts, Screws and Nuts*; IS 2693: *Cast Iron Flange Couplings*.
6. Open Cascade Technology, OCCT geometric modelling kernel, and CadQuery 2.8 parametric modelling library.
7. FreeCAD, open source parametric CAD application. TechDraw workbench, used for projection and sheet generation.
8. Dhondt, G., CalculiX: a three dimensional structural finite element program, used with the gmsh mesh generator.

APPENDIX A SCREENSHOTS OF THE WORKING SYSTEM

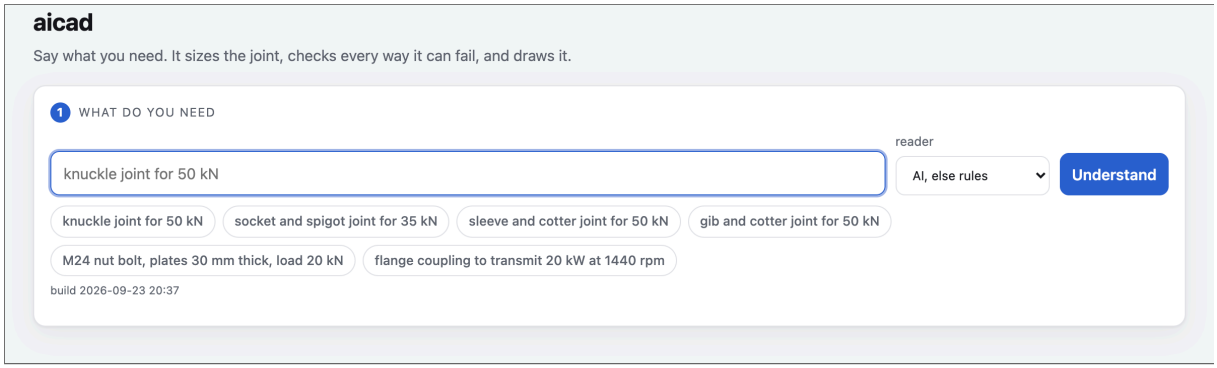


Fig. 4: The request screen. A requirement is typed in ordinary English, or one of the example requests is selected

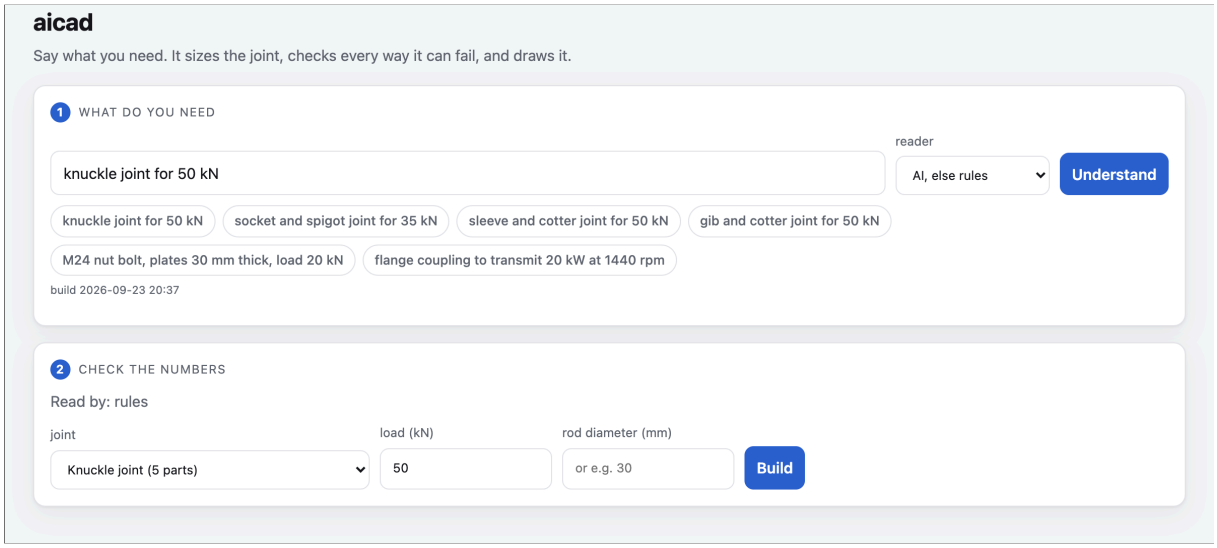


Fig. 5: Parameters extracted from the request, presented for confirmation or correction before the design proceeds

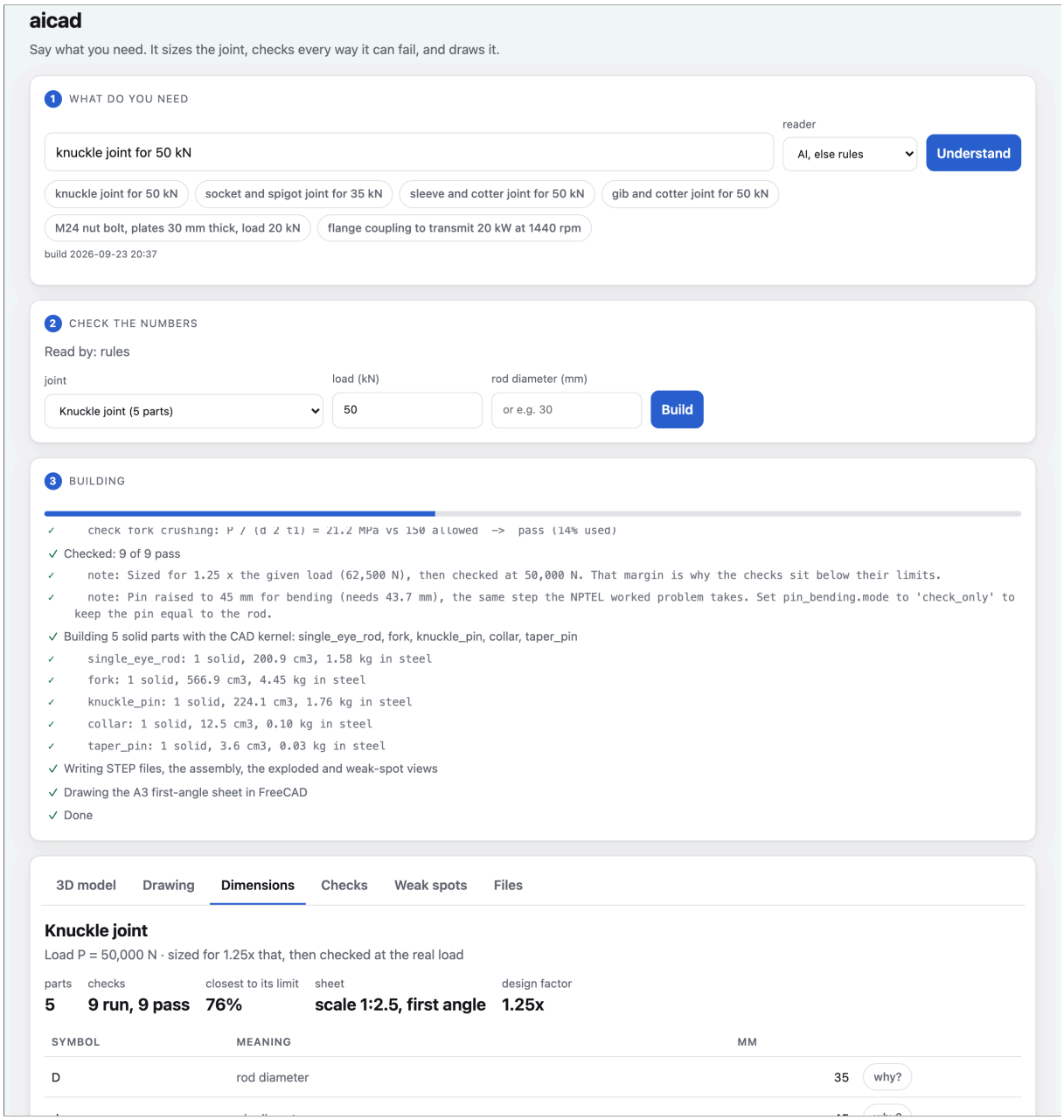


Fig. 6: Every computed dimension with the rule that produced it and its source; a plain language explanation is available for each

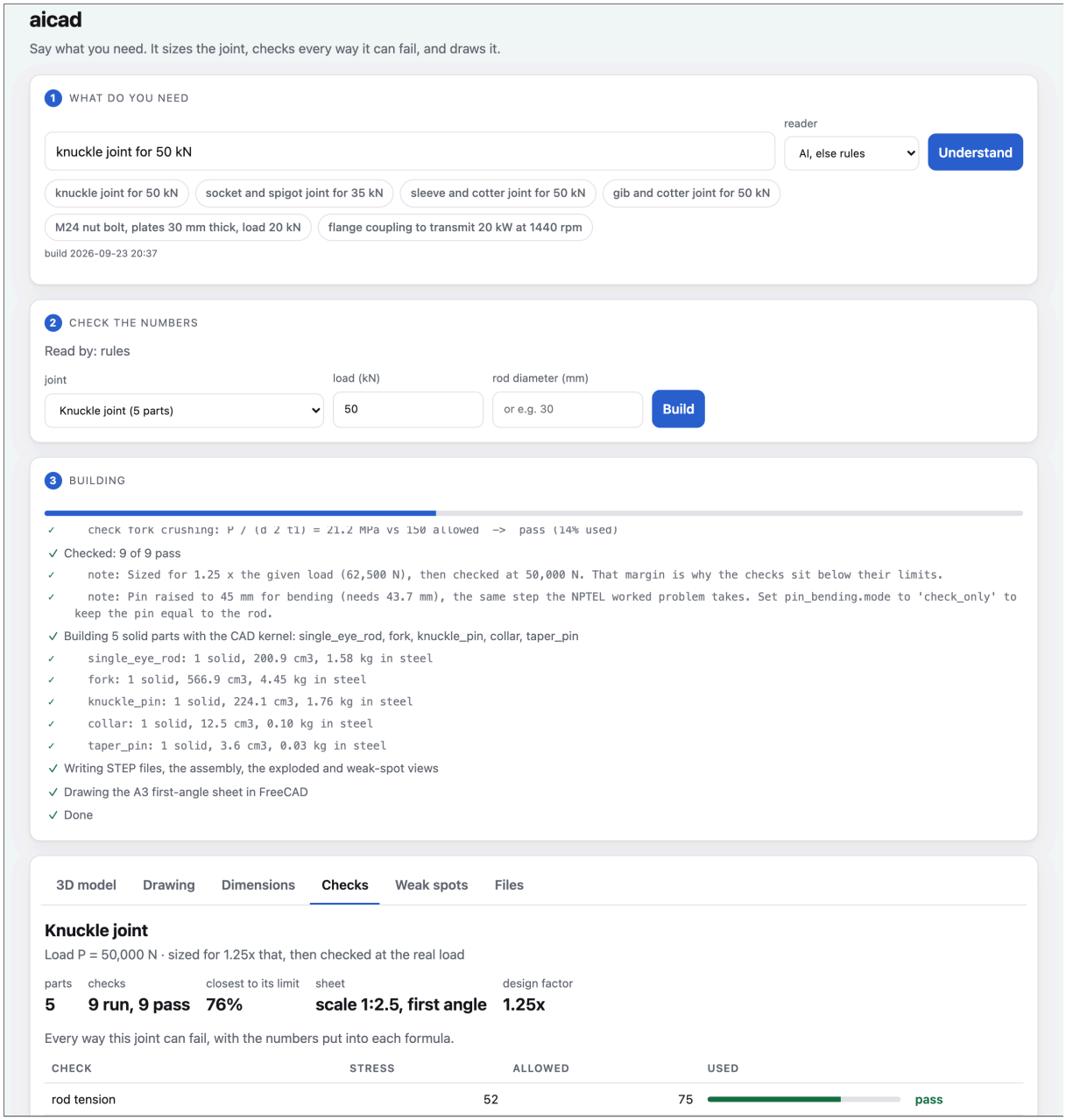


Fig. 7: Every failure check with its substituted arithmetic, its allowable stress and its utilisation

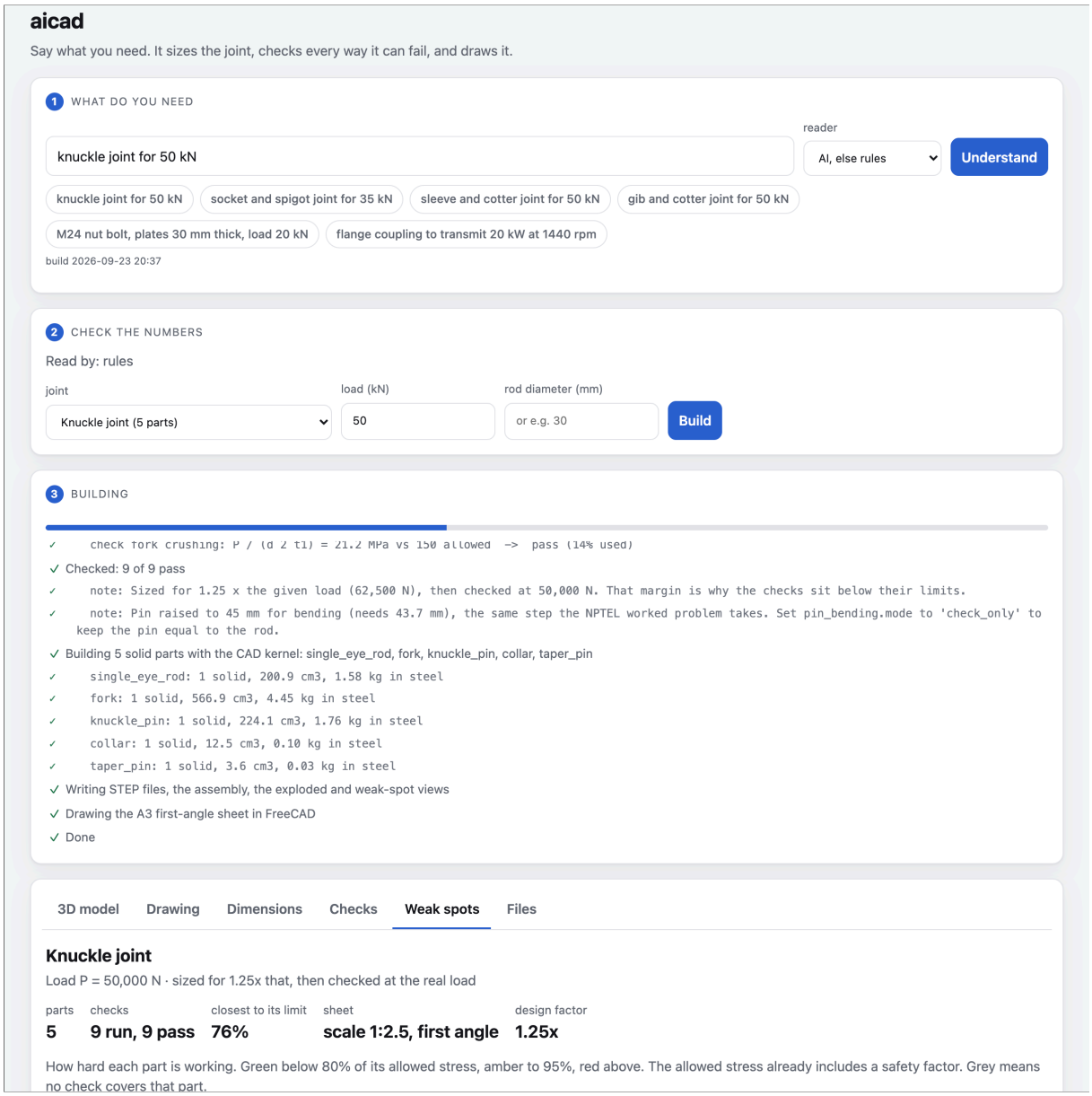


Fig. 8: Components coloured by utilisation, so that the member nearest its limit is apparent without reading the table

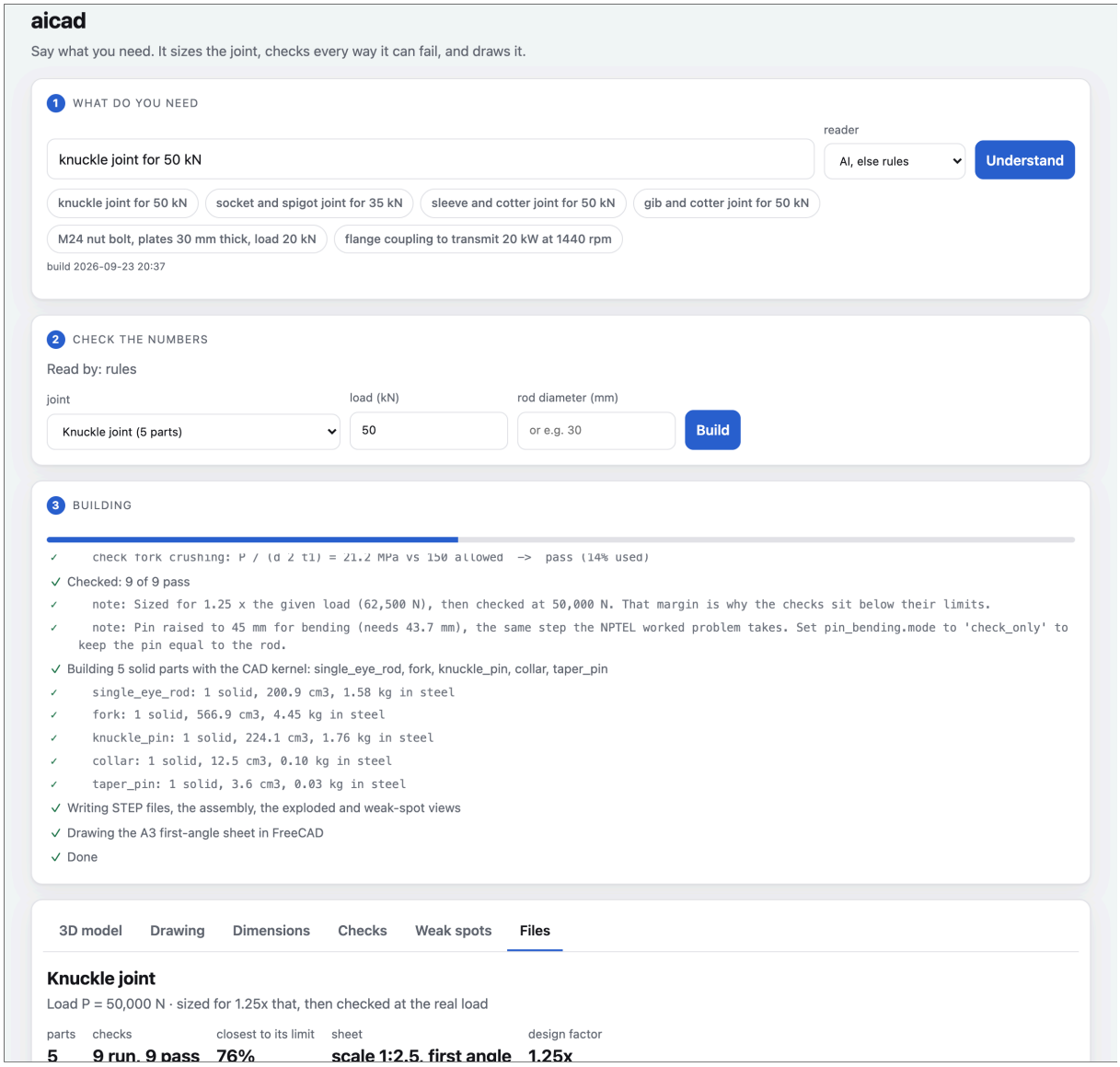


Fig. 9: Files produced by a single run: a STEP file for each component, the assembly, the drawing sheet, the DXF, the FreeCAD file and a written report